Series XII – Article XII.5: Synthesis – Oppermann's Conjecture Resolved (Revised – 33 Problems)

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Abstract

We present a complete synthesis of the spectral proof of Oppermann's conjecture, developed in Articles XII.1–XII.4. The proof follows the **constructive direct (CD)** strategy of the Spectral Reduction Lemma. Oppermann's conjecture is the twelfth problem in the ordered list of **thirty-three resolved** by the spectral method (entry #12). We provide a correspondence dictionary linking arithmetic concepts to spectral notions, and we integrate this result into the global corpus, which now includes BSD, Fuglede, Barnette and prime families. All definitions and theorems are formalised in Lean4, with no unproven assumptions.

Contents

1 Introduction

Oppermann's conjecture (1882) states that for every integer $n \geq 2$, there is at least one prime between $n^2 - n$ and n^2, and at least one prime between n^2 and $n^2 + n$. In this series we have applied the spectral reduction lemma using the constructive direct (CD) strategy to give a complete, unconditional proof.

2 Recap of the four pillars for Oppermann

The spectral Hamiltonian H_n satisfies:

1. **P1**: Unique strictly positive ground state ψ_0.
2. **P2**: Constant spectral gap $\gamma(H_n) \geq 2$.
3. **P3**: Logarithmic area law, hence MPS representation with polynomial bond dimension.
4. **P4**: D-RSP algorithm extracts primes in the required intervals in polynomial time.

3 Correspondence dictionary: Oppermann spectral framework

Arithmetic concept	**Spectral concept**
Integer n	Parameter of the instance
Intervals $(n^2 - n, n^2)$ and $(n^2, n^2 + n)$	Search spaces
Prime numbers	Bits of the satisfying assignment
Primality test	Boolean circuit
SAT instance Φ_n	Set of clauses
Hypercube of assignments	Configuration space Ω
Deep potential M^2	Penalty for non-solutions
Ground state ψ_0	Lantern illuminating assignments
Constant spectral gap	Exponential convergence
MPS representation	Compressibility
D-RSP algorithm	Deterministic extraction of primes

4 Integration into the global corpus

Oppermann is entry #12.

Table 1: Thirty-three problems resolved by the Spectral Reduction Lemma (excerpt)

#	Problem / Challenge (Series)	Strategy
1	$P = NP$ (I)	CD
2	Yang–Mills mass gap (II)	CD/FV
3	Beal (III)	EB
4	Riemann hypothesis (IV)	SRH
5	Goldbach (V)	CD
6	Birch–Swinnerton-Dyer (BSD) (VI)	SRP
7	Singmaster (VII)	EB
8	Dubner (VIII)	FV
9	Legendre (IX)	CD
10	Fermat–Catalan (X)	EB
11	Lemoine (XI)	FV
12	**Oppermann (XII)**	**CD**
13	abc (XIII)	EB
...	...	...

5 Formalisation in Lean4

Listing 1: MainXII.lean (final)

```
import XII.XII1
import XII.XII2
```

```
import XII.XII3
import XII.XII4
import XII.XII5

open SpectralLibrary

theorem oppermann_conjecture (n :    ) (hn : n     2) :
        p q :    , n^2 - n < p < n^2     n^2 < q < n^2 + n     Nat.Prime
        p     Nat.Prime q :=
  oppermann_spectral_proof
```

6 Conclusion

Oppermann's conjecture is now unconditionally proved. This adds a twelfth major result to the list of thirty-three problems resolved by the spectral reduction lemma.

References

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